**CHAPTER 08**

# Carboxylic Acids

## Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

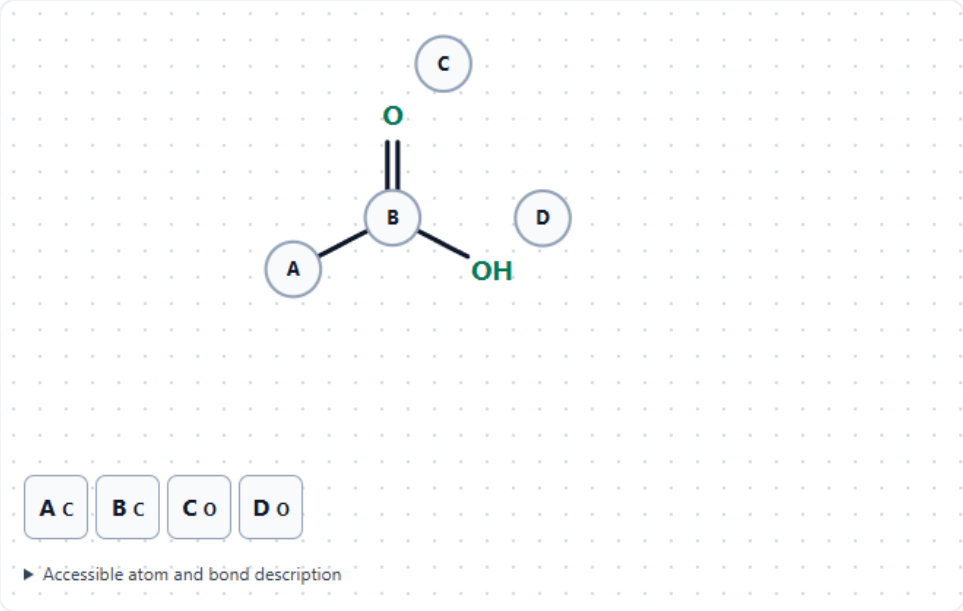
Source snapshot SHA-256: a5681a8efd4fd7f05ab6ac941f057bbfd23a13c1196fc5435c9fced0ba6c2485

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

# acid-ratio | Independent exercise

Screenshot: acid-ratio-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## acid-ratio:stage-1 | Select the complete carboxyl group: its carbon and both oxygens.

Answer type: atoms

## acid-ratio:stage-2 | For pKa 4.8 at pH 6.8, what is [A−]/[HA]?

Answer type: number | unit: dimensionless

## acid-ratio:stage-3 | Which form predominates?

Answer type: choice

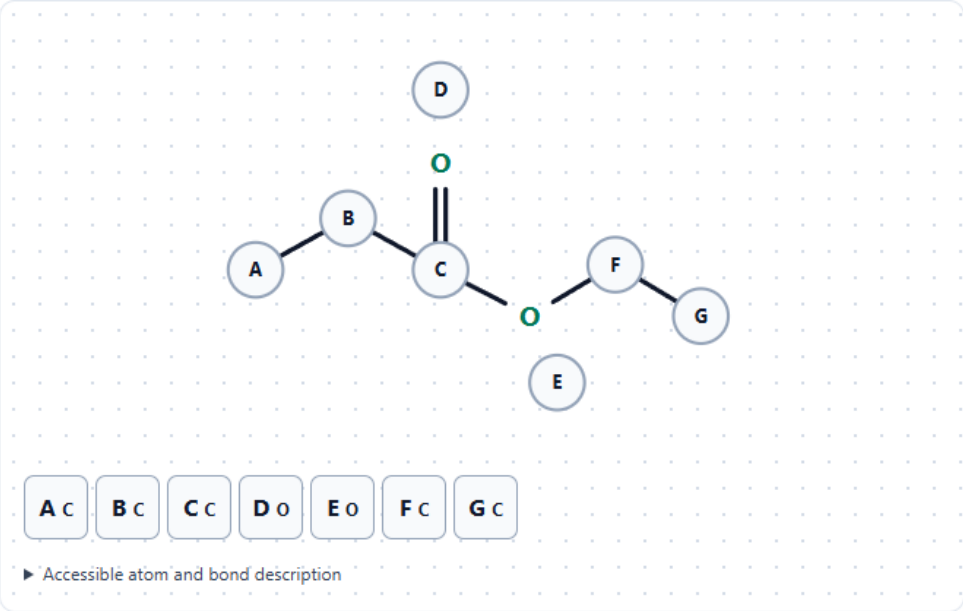
Option 0 (A): Neutral acid

Option 1 (B): Carboxylate

# make-ester | Independent exercise

The ester canvas illustrates the functional-group pattern. The later acid/amine question is a separate transfer question, not a transformation of this ester.

Screenshot: make-ester-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

## make-ester:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

## make-ester:stage-2 | Removing water from a Fischer esterification generally shifts toward...

Answer type: choice

Option 0 (A): Ester

Option 1 (B): Acid and alcohol

Option 2 (C): No equilibrium change

## make-ester:stage-3 | Mixing a carboxylic acid and an amine in water initially favors...

Answer type: choice

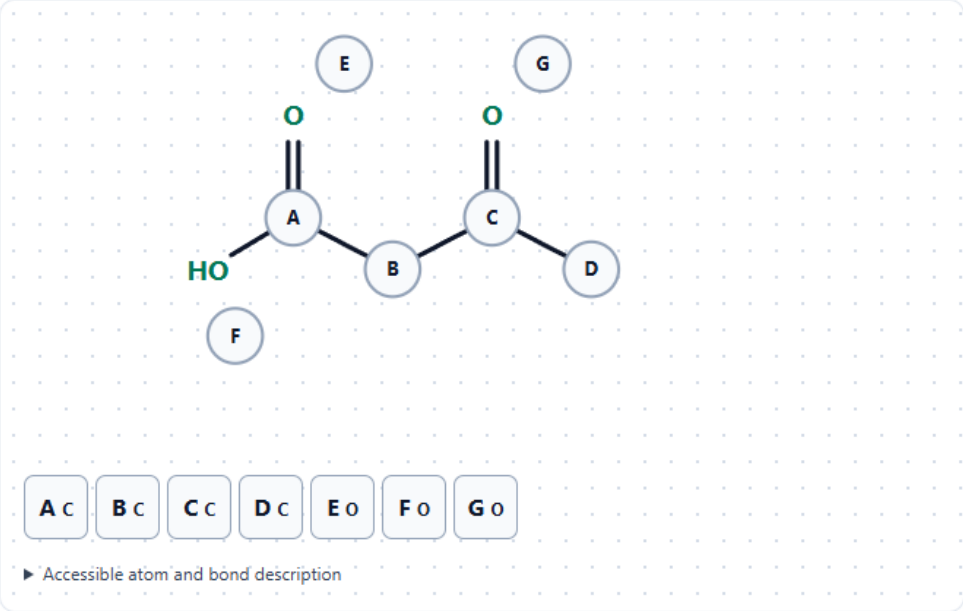
Option 0 (A): Ammonium carboxylate salt

Option 1 (B): An amide without activation

Option 2 (C): An alkane

# lose-co2 | Independent exercise

Screenshot: lose-co2-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=0.00, y=115.00; charge=0; explicit H override=none

6 (G): O; x=166.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [0, 4, 2], [0, 5, 1], [2, 6, 2]]

## lose-co2:stage-1 | Select the carbon lost as CO2.

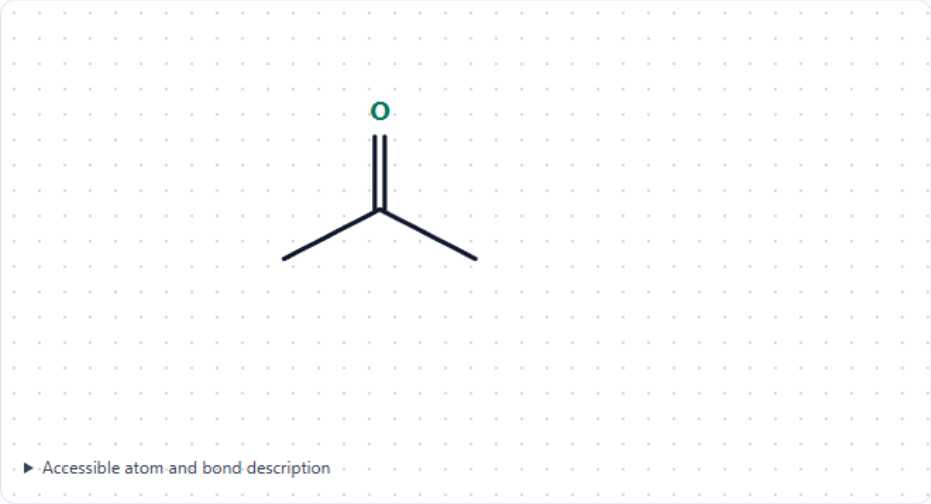
Answer type: atoms

## lose-co2:stage-2 | How many carbons remain in the organic product?

Answer type: number | unit: carbons

This stage uses a different molecule (connectivity shown).

Screenshot: lose-co2-stage-3-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

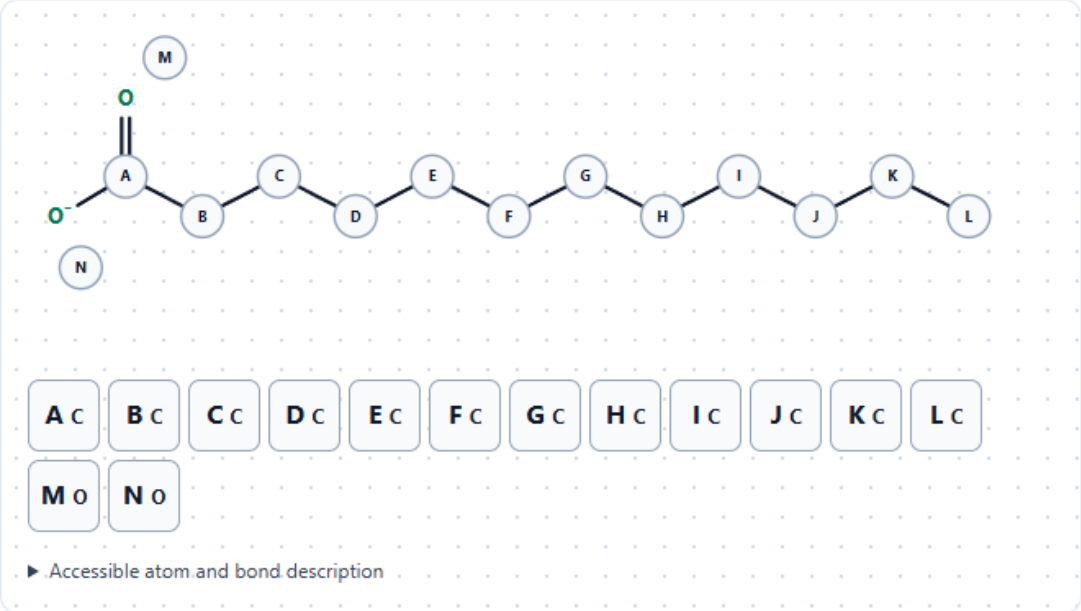
Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

## lose-co2:stage-3 | Name the carbonyl product after tautomerization.

Answer type: text

# soap-head | Independent exercise

Screenshot: soap-head-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): C; x=282.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=340.00, y=115.00; charge=0; explicit H override=none

6 (G): C; x=398.00, y=85.00; charge=0; explicit H override=none

7 (H): C; x=456.00, y=115.00; charge=0; explicit H override=none

8 (I): C; x=514.00, y=85.00; charge=0; explicit H override=none

9 (J): C; x=572.00, y=115.00; charge=0; explicit H override=none

10 (K): C; x=630.00, y=85.00; charge=0; explicit H override=none

11 (L): C; x=688.00, y=115.00; charge=0; explicit H override=none

12 (M): O; x=50.00, y=25.00; charge=0; explicit H override=none

13 (N): O; x=0.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 1], [7, 8, 1], [8, 9, 1], [9, 10, 1], [10, 11, 1], [0, 12, 2], [0, 13, 1]]

## soap-head:stage-1 | Select the two oxygen atoms of the ionic head.

Answer type: atoms

## soap-head:stage-2 | In an aqueous micelle, hydrocarbon tails tend to point…

Answer type: choice

Option 0 (A): Toward the interior

Option 1 (B): Out into surrounding water

## soap-head:stage-3 | Saponification of a triacylglycerol produces…

Answer type: choice

Option 0 (A): Glycerol and fatty-acid salts

Option 1 (B): Only free neutral fatty acids under strongly basic conditions

Option 2 (C): A protein